Chapter 3: Discrete Random Variables and Probability Distributions

PART A:

- RANDOM VARIABLES
- DISCRETE PROBABILITY DISTRIBUTION

Learning Objectives:

At the end of this chapter, the students would be able to:

- ✓ Understand the concept of a discrete random variable and probability distribution
- ✓ Be able to compute and interpret the expected value, variance, and standard deviation for a discrete random variable
- ✓ Be able to compute probabilities using a binomial probability distribution
- ✓ Be able to compute probabilities using a Poisson probability distribution

3.1 Concept of Random Variable

$P(X=3) = \frac{1}{5}$

$P(X=x) = x_1, x_2, x_3$

- **Random variable:** A function that assigns numerical values to the outcomes of an experiment.

- Denoted by uppercase letters (e.g., *X* or *Y*).

- Values of the random variable are denoted by corresponding lowercase letters: $x_1, x_2, x_3, \ldots, x_n$

3.1 Concept of Random Variable

- Notation:

$$P(X = x)$$

"Probability that the random variable *X* takes the value of *x*"

OR

"The probability that *X* has the value *x*"

3.1 Concept of Random Variable

- Example:

Experiment	Random Variable	Value
Flip two coins once	X = the total number of tails	x = 0, 1, 2
Flip a coin three times	Y = the total number of heads	y_1 y_2 y_3 y_4 y = 0, 1, 2, 3
Throw two dice	Z = the sum of the numbers facing up	Z = 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12

3.1 Concept of Random Variable

- Random variables may be classified as:

  - **Discrete:** The random variable assumes a countable number of distinct values.

  - **Continuous:** The random variable is characterized by uncountable values in an interval.

(A) Discrete Random Variables

- Discrete random variables can only take exact/countable values

0, 1, 2, 3, 4,…

- Consider an experiment in which two apples are selected from the production line and each is either rotten (R) or non-rotten (N). The sample space as shown in Figure 1.
- The random variable *X* is the number of rotten apples.
- The possible number of rotten apple is the set *X* = {0, 1, 2}.
- Since these are the only possible outcomes, *X* is a **discrete random variable**.

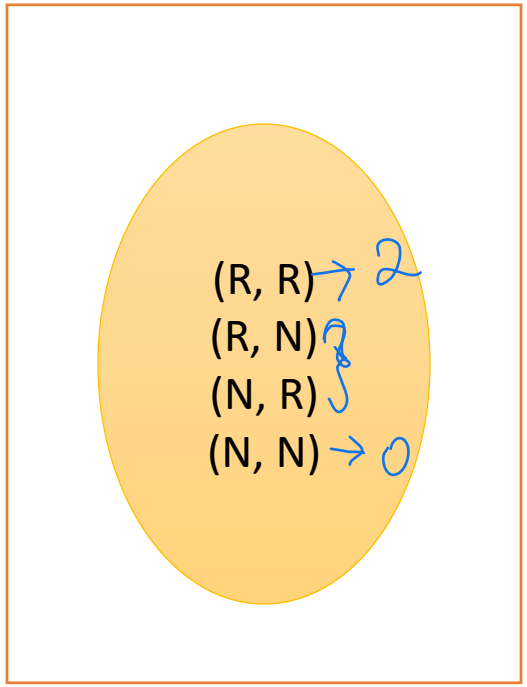


Figure 1

(A) Discrete Random Variables

- A **discrete probability distribution** is a table which lists all the possible values of a discrete random variable with their corresponding probabilities
- Key properties of discrete probability distributions:

❑ Its probability mass function *f(x)* is specified as:

$$P(X = x) = f(x)$$

… for all *x* in the range of *X*.

(A) Discrete Random Variables

- Key properties of discrete probability distributions:

❑ The probability of each value *x* is a value between 0 and 1, or equivalently:

$$0 \leq P(X = x) \leq 1$$

❑The sum of the probabilities equals 1. In other words:

$$\sum P(X = x) = 1$$

or

$$\sum f(x) = 1$$

where the sum extends over all values x of X.

(A) Discrete Random Variables

- *X* can be in **probability distribution table** form.
- For example, consider the experiment of rolling a six-sided die. The probability distribution is:

X	**1**	**2**	**3**	**4**	**5**	**6**
P(*X* = *x*)	1/6	1/6	1/6	1/6	1/6	1/6

- Each outcome has an associated probability of 1/6. Thus, the pairs of values and their probabilities form the **probability mass function** for *X*.

(A) Discrete Random Variables

- *X* can be in the form of **probability function** form.

$$f(x) = \begin{cases} \frac{1}{8} \; if \; x = 0,3 \\ \frac{3}{8} \; if \; x = 1,2 \\ 0 \quad otherwise \end{cases}$$

f(0)

f(2)

f(6) = 0

Simple exercise:

Consider the probability distribution that reflects the number of motorcycle sold by a salesperson over a one-week period:

Number of motorcycle sold	Probability
0	0.30
1	0.50
2	0.15
3	0.05

(i) What is the probability that salesperson does not sell any motorcycle?

(ii) What is the probability that the salesperson sells at most one motorcycle?

(iii) What is the probability that the realtor sells at least two houses?

$P(X \leq 1) = P(X=0) + P(X=1)$
$= 0.3 + 0.5$
$= 0.8$

x $P(X=x)$ $x \cdot P(X=x)$

$P(X=0) = 0.3$

0
0.5
0.3
0.15

$= 0.20$

$P(X \geq 2)$

$E(X) = 0.95$

Expected value, Variance, and Standard deviation

Summary measures for a random variable include the

- ✓ Mean (Expected Value)
- ✓ Variance
- ✓ Standard Deviation

Expected value

$E(3X-1) = E(3X) - E(1)$
$= 3E(X) - 1$
$= 3(0.95) - 1$
$=$

$E(4) = 4$

The expected value (mean) or average of random variable defined by:

- $E(X) = \sum x \,.\, P(X = x)$
- If a is constant, $E(a) = a$
- If a and b are constants, then $E(aX \pm b) = aE(X) \pm b$

Expected value

Exercise (from textbook Introduction to Probability and Statistics, UUM, (2021); pg. 63)

x	0	1	2	3	Total
$f(x) = P(x = X)$	1/8	3/8	3/8	1/8	1

$x \cdot P(x=x) = (0 \times \frac{1}{8}) + (1 \times \frac{3}{8}) + (2 \times \frac{3}{8}) + (3 \times \frac{1}{8}) = 1.5$

Calculate:

(i) $E(X)$

(ii) $E(X^2)$ $(0^2 \times \frac{1}{8}) + (1^2 \times \frac{3}{8}) + (2^2 \times \frac{3}{8}) + (3^2 \times \frac{1}{8}) = 3$

(iii) $E(4X)$ $4(E(x)) = 4 \times 1.5 = 6$ *

(iv) $E(4X - 2)$ $= E(4X) - E(2)$

$= 4(E(X)) - 2$

$= 6 - 2$

$= 4$ *

Variance, and Standard deviation

- Formula for Variance (Var) and Standard deviation (SD):

$$Var(X) = \sigma^2 = \sum (x_i - \mu)^2 \, P(X = x_i)$$

$$Var\,(X) = E\left(X^2\right) - E(X)^2$$

$$Var\,(X) = E\left(X^2\right) - \mu^2$$

$E(X)$; the mean

$$SD(X) = \sigma = \sqrt{\sigma^2}$$

$$SD(X) = \sqrt{Var(X)}$$

Variance, and Standard deviation

$E(8) = 8$

- If a is constant, then:

$$Var\,(a) = 0$$

$$\text{SD}\,(a) = 0$$

$var(8) = 0$

$var(8X - 1)$

$= var(8X) - var(1)$

$= 8^2 var(X)$ $\quad 0$

- If X is a random variable, with a and b are constants then:

$$Var\,(aX \pm b) = a^2 Var(X)$$

$$SD\,(aX \pm b) = |a| SD(X)$$

$P(X=0) + P(X=1) + P(X=2) = \frac{7}{8}$

Variance and standard deviation

$\frac{3}{8} + \frac{1}{8} = \frac{4}{8} = \frac{1}{2}$

Exercise (from textbook Introduction to Probability and Statistics, UUM, (2021); pg. 63)

x	0	1	2	3	Total
$f(x) = P(x = X)$	1/8	3/8	3/8	1/8	1

Calculate:

(i) $Var(X)$ → $E(X^2) - (E(X))^2 = 3 - (1.5)^2 = 0.75$

(ii) $Var(4X - 1)$ → $= var(4X) - var(1)$ → $= 4^2(var(X))$ → $16(0.75) = 12$

(iii) $Var(7X + 3)$ → $= var(7X) + var(3)$ → $= 7^2(0.75)$ → $= 36.75$

(iv) $SD(X)$ → $= \sqrt{var(X)} = \sqrt{0.75} = 0.866$

(v) $P(X < 3)$

(vi) $P(X \geq 2)$

Expected value, Variance, and Standard deviation (EXTRA EXAMPLE)

Example: Brad Williams, owner of a car dealership in Chicago, decides to construct an incentive compensation program based on performance.

Bonus (in $1,000s)	Performance Type	Probability
10	Superior	0.15
6	Good	0.25
3	Fair	0.40
0	Poor	0.20

- ✓ Calculate the expected value of the annual bonus amount.
- ✓ Calculate the variance and standard deviation of the annual bonus amount.

Expected value, Variance, and Standard deviation (EXTRA EXAMPLE)

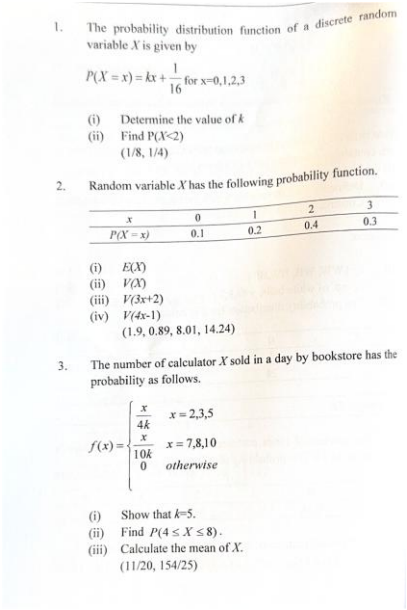
Solution: Let the random variable X denote the bonus amount (in \$1,000s) for an employee.

Value, x_i	Probability, $P(X = x_i)$	Weighted Value, $x_i P(X = x_i)$	Weighted Squared Deviation, $(x_i - \mu)^2 P(X = x_i)$
10	0.15	$10 \times 0.15 = 1.5$	$(10 - 4.2)^2 \times 0.15 = 5.05$
6	0.25	$6 \times 0.25 = 1.5$	$(6 - 4.2)^2 \times 0.25 = 0.81$
3	0.40	$3 \times 0.40 = 1.2$	$(3 - 4.2)^2 \times 0.40 = 0.58$
0	0.20	$0 \times 0.20 = 0$	$(0 - 4.2)^2 \times 0.20 = 3.53$
		Total = 4.2	Total = 9.97

- ✓ $E(X) = \mu = \Sigma\, x_i P(X = x_i) =$ **4.2**
- ✓ $Var(X) = \sigma^2 = \Sigma\, (x_i - \mu)^2 P(X = x_i) =$ **9.97**
- ✓ $SD(X) = \sigma =$ **3.16**

EXERCISES

Exercise 3.2 from textbook Introduction to Probability and Statistics, UUM, (2021); pg. 66 – 67



1. The probability distribution function of a discrete random variable X is given by

$$P(X=x)=kx+\frac{1}{16} \text{ for } x=0,1,2,3$$

(i) Determine the value of k
(ii) Find $P(X<2)$
(1/8, 1/4)

2. Random variable X has the following probability function.

x	0	1	2	3
$P(X=x)$	0.1	0.2	0.4	0.3

(i) $E(X)$
(ii) $V(X)$
(iii) $V(3x+2)$
(iv) $V(4x-1)$
(1.9, 0.89, 8.01, 14.24)

3. The number of calculator X sold in a day by bookstore has the probability as follows.

$$f(x)=\begin{cases}\frac{x}{4k} & x=2,3,5\\ \frac{x}{10k} & x=7,8,10\\ 0 & otherwise\end{cases}$$

(i) Show that $k=5$.
(ii) Find $P(4\le X\le 8)$.
(iii) Calculate the mean of X.
(11/20, 154/25)

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